

Aim: Create a simple Spark MLlib pipeline to classify data. Use logistic regression or decision trees.

Code:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, when
from pyspark.ml import Pipeline
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import BinaryClassificationEvaluator
```

```
spark = SparkSession.builder \
    .appName("SparkMLlibClassification") \
    .master("local[*]") \
    .getOrCreate()
```

```
print("=====")
print("    SPARK MLlib CLASSIFICATION")
print("    Shreyash Kadam - S091")
print("=====")
```

```
data = spark.read.csv(
    "all_stocks_5yr.csv",
    header=True,
    inferSchema=True
)
```

```
print("\nOriginal Dataset:")
data.show(10)
```

```
data = data.select(
    "open",
    "high",
    "low",
    "close",
    "volume"
)
```

```
data = data.dropna()
```

```
data = data.withColumn(
    "label",
    when(col("close") > col("open"), 1.0)
    .otherwise(0.0)
```

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```
)

print("\nDataset with Label:")
data.show(10)

print("\nClass Distribution:")
data.groupBy("label").count().show()

features = [
    "open",
    "high",
    "low",
    "volume"
]

assembler = VectorAssembler(
    inputCols=features,
    outputCol="features"
)

lr = LogisticRegression(
    featuresCol="features",
    labelCol="label",
    predictionCol="prediction"
)

pipeline = Pipeline(
    stages=[
        assembler,
        lr
    ]
)

train_data, test_data = data.randomSplit(
    [0.8, 0.2],
    seed=42
)

print("\nTraining Data:", train_data.count())
print("Testing Data :", test_data.count())

model = pipeline.fit(train_data)

predictions = model.transform(test_data)
```

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```

print("\nPredictions:")
predictions.select(
    "open",
    "high",
    "low",
    "volume",
    "label",
    "prediction"
).show(20)

true_positive = predictions.filter(
    (col("label") == 1.0) &
    (col("prediction") == 1.0)
).count()

true_negative = predictions.filter(
    (col("label") == 0.0) &
    (col("prediction") == 0.0)
).count()

false_positive = predictions.filter(
    (col("label") == 0.0) &
    (col("prediction") == 1.0)
).count()

false_negative = predictions.filter(
    (col("label") == 1.0) &
    (col("prediction") == 0.0)
).count()

print("\n=====")
print("          CONFUSION MATRIX")
print("=====")

print("          Predicted")
print("          0      1")
print("Actual 0      ", true_negative, " ", false_positive)
print("Actual 1      ", false_negative, " ", true_positive)

total = (
    true_positive +
    true_negative +
    false_positive +

```

```
        false_negative
    )

    accuracy = (
        (true_positive + true_negative) / total
        if total > 0 else 0
    )

    sensitivity = (
        true_positive / (true_positive + false_negative)
        if (true_positive + false_negative) > 0
        else 0
    )

    precision = (
        true_positive / (true_positive + false_positive)
        if (true_positive + false_positive) > 0
        else 0
    )

    f1_score = (
        2 * precision * sensitivity /
        (precision + sensitivity)
        if (precision + sensitivity) > 0
        else 0
    )

    auc_evaluator = BinaryClassificationEvaluator(
        labelCol="label",
        rawPredictionCol="rawPrediction",
        metricName="areaUnderROC"
    )

    auc = auc_evaluator.evaluate(predictions)

    print("\n=====")
    print("      CLASSIFICATION RESULTS")
    print("=====")

    print(f"Accuracy    : {accuracy:.4f}")
    print(f"Sensitivity : {sensitivity:.4f}")
    print(f"Precision   : {precision:.4f}")
    print(f"F1 Score    : {f1_score:.4f}")
    print(f"AUC        : {auc:.4f}")
```

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```
print("\n=====")  
print("          END")  
print("=====")  
  
spark.stop()
```

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Output:

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```
=====
      SPARK MLlib CLASSIFICATION
      Shreyash Kadam - S091
=====

Original Dataset:
+-----+-----+-----+-----+-----+-----+
|   date| open| high|  low|close|  volume|Name|
+-----+-----+-----+-----+-----+-----+
|2013-02-08|15.07|15.12|14.63|14.75| 8407500| AAL|
|2013-02-11|14.89|15.01|14.26|14.46| 8882000| AAL|
|2013-02-12|14.45|14.51| 14.1|14.27| 8126000| AAL|
|2013-02-13| 14.3|14.94|14.25|14.66|10259500| AAL|
|2013-02-14|14.94|14.96|13.16|13.99|31879900| AAL|
|2013-02-15|13.93|14.61|13.93| 14.5|15628000| AAL|
|2013-02-19|14.33|14.56|14.08|14.26|11354400| AAL|
|2013-02-20|14.17|14.26|13.15|13.33|14725200| AAL|
|2013-02-21|13.62|13.95| 12.9|13.37|11922100| AAL|
|2013-02-22|13.57| 13.6|13.21|13.57| 6071400| AAL|
+-----+-----+-----+-----+-----+-----+
only showing top 10 rows

Dataset with Label:
+-----+-----+-----+-----+-----+-----+
| open| high|  low|close|  volume|label|
+-----+-----+-----+-----+-----+-----+
|15.07|15.12|14.63|14.75| 8407500| 0.0|
|14.89|15.01|14.26|14.46| 8882000| 0.0|
|14.45|14.51| 14.1|14.27| 8126000| 0.0|
| 14.3|14.94|14.25|14.66|10259500| 1.0|
|14.94|14.96|13.16|13.99|31879900| 0.0|
|13.93|14.61|13.93| 14.5|15628000| 1.0|
|14.33|14.56|14.08|14.26|11354400| 0.0|
|14.17|14.26|13.15|13.33|14725200| 0.0|
|13.62|13.95| 12.9|13.37|11922100| 0.0|
|13.57| 13.6|13.21|13.57| 6071400| 0.0|
+-----+-----+-----+-----+-----+-----+
only showing top 10 rows

Class Distribution:
+-----+-----+
|label| count|
+-----+-----+
| 0.0|300059|
| 1.0|318970|
+-----+-----+
```

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Training Data: 495292

Testing Data : 123737

netlib-blas: JNI_OnLoad: dlopen(libopenblas.dll) failed: LoadLibrary failed

Predictions:

```
+-----+-----+-----+-----+-----+-----+
|open|  high| low|  volume|label|prediction|
+-----+-----+-----+-----+-----+-----+
|1.67|   1.7|1.61|11916468| 0.0|      0.0|
|1.69|   1.73|1.66|10715268| 1.0|      1.0|
| 1.7|   1.72|1.68| 5739499| 1.0|      1.0|
|1.76|   1.77|1.65|16478376| 0.0|      0.0|
|1.77|   1.82|1.71| 8470710| 0.0|      0.0|
| 1.8|   1.81|1.75|10641876| 0.0|      0.0|
|1.81|   1.95|1.75|29243601| 0.0|      1.0|
|1.82|   1.87|1.82| 7859214| 1.0|      1.0|
|1.87|   1.9|1.74|28127072| 0.0|      0.0|
|1.87|   1.9|1.85| 4928291| 1.0|      1.0|
|1.87|   1.92|1.85|11814043| 0.0|      1.0|
|1.87|1.9475|1.85| 9198667| 1.0|      1.0|
|1.88|   2.14|1.81|31010289| 1.0|      1.0|
| 1.9|   1.94|1.87| 5831567| 0.0|      1.0|
|1.92|   1.98|1.83| 9466130| 0.0|      0.0|
|1.96|   1.96|1.85| 7687701| 0.0|      0.0|
|1.99|   2.15|1.98| 9384658| 1.0|      1.0|
|2.08|   2.18|2.07|10833249| 1.0|      1.0|
|2.13|   2.19|2.11| 5858746| 1.0|      1.0|
|2.14|   2.25|2.13|10727076| 1.0|      1.0|
+-----+-----+-----+-----+-----+-----+
```

only showing top 20 rows

=====

CONFUSION MATRIX

=====

	Predicted	
	0	1
Actual 0	49898	10188
Actual 1	9598	54053

=====

CLASSIFICATION RESULTS

=====

Accuracy : 0.8401

Sensitivity : 0.8492

Precision : 0.8414

F1 Score : 0.8453

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```
|1.88| 2.14|1.81|31010289| 1.0| 1.0|
| 1.9| 1.94|1.87| 5831567| 0.0| 1.0|
|1.92| 1.98|1.83| 9466130| 0.0| 0.0|
|1.96| 1.96|1.85| 7687701| 0.0| 0.0|
|1.99| 2.15|1.98| 9384658| 1.0| 1.0|
|2.08| 2.18|2.07|10833249| 1.0| 1.0|
|2.13| 2.19|2.11| 5858746| 1.0| 1.0|
|2.14| 2.25|2.13|10727076| 1.0| 1.0|
+---+-----+---+-----+---+-----+
only showing top 20 rows

=====
CONFUSION MATRIX
=====
               Predicted
               0      1
Actual 0      49898   10188
Actual 1      9598    54053

=====
CLASSIFICATION RESULTS
=====
Accuracy      : 0.8401
Sensitivity    : 0.8492
Precision     : 0.8414
F1 Score      : 0.8453
AUC           : 0.9081

=====
END
=====
SUCCESS: The process with PID 4388 (child process of PID 14892) has been terminated.
SUCCESS: The process with PID 14892 (child process of PID 2916) has been terminated.
SUCCESS: The process with PID 2916 (child process of PID 10044) has been terminated.

Process finished with exit code 0
```